Patentability Report

Patentability Report: Collapsible Bicycle Helmet with Twist-Lock Honeycomb Liner

Date: 2026-10-08 **Researcher:** Hermes Agent **Status:** Directional read — not a patentability opinion **Version:** 4 (final — comprehensive + twist deep-dive)

Invention Summary

A collapsible bicycle helmet comprising:

1. **Honeycomb protective liner** made of overlapping thermoplastic cells
2. **Outer shell + inner honeycomb liner** (two-part construction)
3. **Quarter-turn twist actuation**: shell rotates ~90° relative to liner → cells lock into rigid expanded configuration
4. **Reverse twist**: cells collapse flat for compact storage
5. **Radial cell arrangement** around the head
6. **Impact absorption** through progressive cell wall crumpling

---

Element-by-Element Prior Art Matrix

#	Element	Status	Closest Reference	Disclosure
1	Honeycomb liner	**Anticipated**	WO2023233224A1, US12268268B2, CN122439960A, CN207285346U, EP4179908A1	Honeycomb protective configuration, radial cell direction,

#	Element	Status	Closest Reference	Disclosure
				staggered layers
2	Overlapping/ staggered cells	**Anticipated**	WO2023233224A1	"layers joined to adjacent layers along **staggered** lines of attachment"
3	Locks rigid when actuated	**Anticipated**	WO2023233224A1, US9743704B2, US11857020	End plates engage to form rigid loop; bosses/ catches lock; locking clip
4	Collapses flat	**Anticipated**	WO2023233224A1, US12268268B2	"collapsed (flattened) form"; "collapsed or folded condition"
5	Impact absorption by cell crush	**Anticipated**	WO2023233224A1 + 15+ papers	"absorbing impact energy through progressive crumpling"
6	**Quarter-turn twist actuation**	**Not found in helmets**	US11857020 (180° rotation, not 90°)	Crown rotates 180° relative to base; different mechanism
7	**Quarter-turn twist mechanism (general)**	**Well-known in mechanical engineering**	US4893978, US6575682, US9200660, US12736079, US20110110716A1, US4653970, US5123795	Quarter-turn fasteners, bayonet locks, twist-lock couplings — ubiquitous

Key Patent References (Ranked by Relevance)

Helmet-Specific Prior Art

Rank	Patent	Title	Overlap
1	**WO2023233224A1** / US12685363	Collapsible helmet, honeycomb, end plates engage to form rigid loop	**Very high** — all elements except twist actuation
2	**US12268268B2** / US12740611	Machine vendible expandable helmet, radial honeycomb matrix	**Very high** — honeycomb + fold/expand + side frames
3	**US11857020**	Collapsible protective helmet, crown rotates 180° relative to base	**High** — rotation between helmet components + locking
4	**US11191316** / US11589634	Collapsible helmet, center + side chassis rotate to collapse	**High** — rotation to collapse + locking mechanism
5	**CN122439960A**	Foldable helmet, honeycomb + airbag auto-unfold	**High** — honeycomb + auto-unfold
6	**US9743704B2** / US9867416B2**	Collapsible helmet, hinged panels, bosses/ catches lock	**Medium** — locking mechanism concept
7	**CN207285346U**	Bicycle helmet with flexible honeycomb matrix	**Medium** — base honeycomb helmet
8	**EP4179908A1**	Cellular liners in foam shell	**Medium** — cellular liner retention
9	**WO2018200414A1** (Fend)	Collapsible helmet	

Rank	Patent	Title	Overlap
			Medium — commercial folding helmet
10	**US10905187B1** (Gwenventions)	Collapsible helmet	**Medium** — commercial folding helmet
11	**GB2567411A**	Folding helmet — interlocking forms on mating surfaces	**Medium** — interlocking mating surfaces
12	**EP3558045NWB1**	Helmet comprising a segmented shell (CPC A42B 3/32)	**Medium** — segmented shell
13	**US12245654**	Foldable helmet with struts, hinges, ratchet mechanism (2025)	**Medium** — strut-based folding, ratchet adjustment
14	**US12685359**	Cellular energy-absorbing structure fastening device (2025)	**Medium** — cellular structure with clamping device
15	**US12458095**	Negative stiffness honeycomb shells for sports helmets (2025)	**Medium** — nylon honeycomb, 90% recovery
16	**US20250221487A1**	Foldable helmet with protector elements (2025)	**Medium** — flexible head-fitting element with protector elements

Helmet Rotation/Locking Mechanisms

Patent	Title	Relevance
CA2561540C	Adjustment mechanism for helmet — sliding shell	Shell sections slide + pivot lock (A42B 3/32)

Patent	Title	Relevance
	sections with pivot lock	
CA2970532A1	Protective helmet systems enabling helmet to rotate independent of head	Shell rotates relative to head/ liner
EP3391765A3	Safety helmet with rotary impact buffering — shell rotates relative to liner	Shell-liner rotation for impact buffering
WO2003026452A1	Locking device for adjustable helmets — cam/lever locking	Adjustable helmet with locking device
US20070056071A1	Firefighter's helmet — inner/outer shell retaining mechanism	Inner/outer shell with retaining mechanism
CA3096994A1	Adjustable helmet assembly — front/back shell with cam locking	Cam locking mechanism for helmet shells
WO2023196826A1	Assembly for releasably connecting inner helmet and outer helmet shell	Inner/outer helmet connection system

Quarter-Turn Twist-Lock Mechanisms (General Mechanical Engineering)

Patent	Title	Relevance
US4893978	Quarter-turn detent twistlock	Square bore + latching shoulders, 90° rotation
US6575682	Quarter turn fastener	90° rotation, cam-stop, blade member
US9200660	Quarter turn fastener	Lock arms + lock ribs, 90° rotation, stop ribs
US12736079	Quarter turn retainer	90° rotation, edge profile for quarter-turn
US20110110716A1	Quarter turn locking mechanism	Bayonet catch, 90° rotation, spinal surgery tool
US4653970		

Patent	Title	Relevance
	Quarter-turn fastener	Spiral cam slot, 1/4 turn locking
US5123795	Quarter turn fastener	Slot + tab, 90° rotation, spring-loaded
US6581986B2	Bayonet locking system for vending machines	90° rotation + axial pull
US9353897	Bayonet fastening for pipeline components	30-50° rotation, latching lug

Design Patents (Folding/Collapsible Helmets)

Patent	Title	Assignee
USD774256S1	Folding helmet	Chris J. Katopis
USD873495S1	Collapsible helmet	Fend Corp.
USD759899S1	Bicycle helmet	—
USD903947S1	Bicycle helmet	—
USD903946S1	Bicycle helmet	—
USD1042978S1	Bicycle helmet	—

Full Claims Analysis: Top 3 Patents

1. WO2023233224A1 (Monopro Ltd) — The Killer Reference

Key claim language:

> "A collapsible helmet deployable for protecting a head of a user, the helmet comprising: an arcuate layered structure openable into a honeycomb protective configuration for encompassing at least an upper part of the head, the arcuate layered structure being formed from layers joined to adjacent layers along staggered lines of attachment so as to define, when opened, a honeycomb structure, wherein the staggered lines of attachment are non-straight lines."

Critical features:

- Honeycomb protective configuration ✓
- Staggered lines of attachment ✓
- Arcuate end plates engage to form rigid loop ✓

- Locking angle > 180° (preferably 220°) ✓
- Collapsed (flattened) form ✓
- Impact absorption through progressive crumpling ✓

What it does NOT disclose:

- Quarter-turn twist actuation (uses manual stretching to engage/disengage end plates)
- Shell rotating relative to liner

2. US12268268B2 (Machine Vendible Expandable Helmet)

Key claim language:

> "A bicycle helmet that fits over a surface of a head of a user, the bicycle helmet comprising: at least one segment of flexible cell structures that form a radial honeycomb matrix movable between a folded condition where each side of the at least one segment is disposed generally parallel and an expanded condition where the radial honeycomb matrix of the at least one segment is configured to be expanded at least partially over the head of the user and arranged radially relative to the surface of the head of the user."

Critical features:

- Radial honeycomb matrix ✓
- Folded ↔ expanded ✓
- Side frames at ends ✓
- Sliding latch for head size adjustment ✓
- "first and second frames rotate to a position beyond parallel in the expanded use position" ✓

What it does NOT disclose:

- Quarter-turn twist actuation (uses chin strap pull to expand)
- Shell rotating relative to liner

3. US11857020 (Coco Worldwide Ltd) — The Rotation Reference

Key claim language:

> "A collapsible protective helmet comprising: an impact-resistant base body configured to encircle the head of a wearer, an impact-resistant crown defining a rim portion, coupling means for coupling the crown to the base body to be movable between a protective position and a stored

> position, and releasable locking means for locking the base body and crown together in each of the two positions of the crown."

Critical features:

- Crown rotates 180° relative to base body ✓
- Ball-and-socket joint for rotation ✓
- Over-centre toggle latch for locking ✓
- Honeycomb structure in base body and/or crown ✓
- Collapsed volume reduction > 40% ✓

What it does NOT disclose:

- Quarter-turn (90°) rotation — uses 180° rotation
- Shell rotating relative to liner — uses crown rotating relative to base body
- Twist-lock mechanism — uses pivot + toggle latch

Non-Patent Literature (OpenAlex/ Crossref, 42 refs)

Key papers:

- "Optimisation of an elastomeric pre-buckled honeycomb helmet liner for advanced impact mitigation" (2023, cited 15×)
- "Viscoelastic circular cell honeycomb helmet liners for reducing head rotation and brain strain in oblique impacts" (2024, cited 14×)
- "Evaluation of a novel bicycle helmet concept in oblique impact testing" (2019, cited 99×)
- "Designing for Sustainability and Safety in Urban Micro-mobility: A Novel Helmet Concept" (2023)
- "Modelling and Analysis of Foldable Motorcycle Helmet for Human Comfort" (2020)
- "A Biomechanical Evaluation of a Novel Airbag Bicycle Helmet Concept" (2021, cited 19×)
- "Helmet Design Based on the Optimization of Biocomposite Energy-Absorbing Liners" (2019, cited 92×)
- "An Overview of the Effectiveness of Bicycle Helmet Designs in Impact Testing" (2021, cited 25×)

RETRACTED (excluded from analysis):

- "Structural improvement of 3D woven honeycomb composite liner" (Research Square, retracted)

The Twist Actuation: Deep Analysis

What I found in helmet patents:

- **US11857020**: Crown rotates **180°** relative to base body (not 90°), using a pivot + locking clip. Different mechanism — the crown flips over, not a twist-lock.
- **US11191316/US11589634**: Side chassis rotate relative to center chassis (not shell vs. liner).
- **WO2023233224A1**: End plates engage by manually stretching the structure (not twisting).
- **US9743704B2**: Panels hinge and snap-lock with bosses/ catches (not twist).
- **CA2561540C**: Shell sections **slide** (not rotate) with pivot lock.
- **EP3391765A3**: Shell rotates relative to liner — but for **impact buffering**, not actuation.
- **CA2970532A1**: Shell rotates relative to head — for **impact decoupling**, not actuation.
- **GB2567411A**: Folding helmet with interlocking mating surfaces — uses hinge + fixings, not twist.

What I found in general mechanical engineering:

- Quarter-turn twist-lock mechanisms are **extremely well-known**:
  - US4893978: "one-quarter turn twistlock fastener with very positive quarter turn latching positions"
  - US6575682: "quarter turn fastener... rotated through an arcuate extent of up to approximately 90°"
  - US9200660: "quarter turn fastener... rotated 90 degrees clockwise"
  - US12736079: "quarter turn retainer... rotated approximately 90 degrees"
  - US20110110716A1: "quarter turn locking mechanisms... bayonet catch"
  - US4653970: "quarter-turn fastener... spiral cam slot... 1/4 turn"
  - US5123795: "quarter turn fastener... one fourth of the way around the circumference"
  - Plus: bayonet couplings, camera lens mounts, twist-lock bottle caps, etc.

What I did NOT find:

- A patent disclosing a **quarter-turn (90°) twist** of the shell relative to the liner to lock/unlock a honeycomb structure **in a helmet**.

But:

- Quarter-turn twist-lock mechanisms are **ubiquitous in mechanical engineering** (dozens of patents, 1970s-2020s)
- Applying a known mechanical principle to a known structure is **obvious** under 35 U.S.C. § 103 (KSR v. Teleflex)
- The US11857020 patent already teaches rotation between helmet components as an actuation mechanism
- An examiner would likely combine US11857020 (helmet rotation) + US4893978/US6575682 (quarter-turn lock) → reject under § 103

Design Patent Landscape

Six design patents found for folding/collapsible helmets:

- **USD774256S1** (Katopis, 2016) — folding helmet
- **USD873495S1** (Fend Corp, 2020) — collapsible helmet
- **USD759899S1**, **USD903947S1**, **USD903946S1**, **USD1042978S1** — bicycle helmets

Design patents protect **ornamental appearance**, not function. If your helmet has a unique visual appearance (e.g., visible twist-lock mechanism, unique honeycomb pattern, distinctive folded shape), a design patent may be viable alongside or instead of a utility patent.

Honest Verdict

Core concept is heavily anticipated. The combination of:

- Honeycomb liner ✓
- Overlapping/staggered cells ✓
- Locks rigid when actuated ✓
- Collapses flat ✓
- Impact absorption by cell crush ✓

...is disclosed in **WO2023233224A1** and the **US12268268B2/ US12740611** family. This is a crowded, commercially active field (Closca, FEND, Overade PLIXI, Newlane all sell folding helmets).

What survives as potentially novel: the specific **quarter-turn twist actuation** — shell rotating 90° relative to liner to switch rigid↔collapsed. No patent found disclosing this exact mechanism in a helmet. However:

1. It's a **narrow claim** at best

2. It's **likely obvious** in view of:
   - US11857020 (rotation between helmet components)
   - US4893978 / US6575682 / US9200660 (quarter-turn twist-lock fasteners — decades of prior art)
   - KSR v. Teleflex (applying known mechanical principle to known structure = obvious)
3. An examiner would likely reject it under § 103 as an obvious combination
4. The **specific geometry** of how the twist actuates the honeycomb cells (cam surfaces, detent positions, etc.) might be the only protectable aspect — but this would be a very narrow design patent or utility patent claim

This is a directional read, not a patentability opinion. For a real filing decision you need a classification-driven search (CPC **A42B 3/125**, **A42B 3/32**, **A42B 3/128**) by a registered patent attorney.

Search Methodology

- **Sources:** web_search (Hermes backend), web_extract (Google Patents), OpenAlex, Crossref
- **Patent77 search:** Rate-limited (daily quota of 20 searches reached)
- **Patent77 get_patent:** Upstream unavailable (rate-limits heavy use)
- **Direct curl:** IP-throttled by Google (503)
- **Queries:** 4 patent framings + 2 literature framings + 2 twist-lock framings + 1 CPC classification + 1 design patent
- **Total unique refs:** 42 NPL + 12 helmet patents + 7 helmet rotation patents + 9 quarter-turn patents + 6 design patents = **76 total**
- **Search date:** 2026-10-08

Commercial Products in the Space

- **Closca** — folding honeycomb commuter helmet (€89.90)
- **Overade PLIXI** — foldable urban bike helmet
- **FEND One** — folding helmet, 50% size reduction
- **Newlane** — collapsible cycling helmet with Flip-Clip
- **HIVE** — origami-inspired collapsible bike helmet (student project)

Recommended Next Steps

1. **Classification-driven search** by a registered patent attorney (CPC A42B 3/125, A42B 3/32, A42B 3/128)
2. **Deep search on twist actuation geometry** — search for specific cam/detent geometries in helmet contexts
3. **Consider design patent** instead of utility patent — the visual appearance of a twist-lock folding helmet may be protectable (6 design patents already exist in this space)
4. **Provisional filing** if you want to preserve a priority date while continuing research
5. **Consider the specific honeycomb geometry** — if your cell geometry is novel (e.g., specific staggered pattern, specific cell shape), that might be independently patentable
6. **Patent77 retry tomorrow** — their daily quota resets; their AI search may find references my keyword search missed

---

Deep-Dive Update (v4): Is the Quarter-Turn Twist Actuation Novel?

Follow-up search (2026-10-09) targeting the only surviving differentiator — shell rotates ~90° relative to the liner to lock the honeycomb rigid. Five query framings were run through the Hermes `web_search` backend (bypasses the throttled device IP), plus direct extraction of the top candidates.

New references found (not in v3)

Ref	Title / Assignee	What it discloses	Impact on twist claim
US11944149B2	Helmet — **MIPS AB**	Inner shell + outer shell + **sliding interface** between them; a **switch** selectively switchable between first/second discrete modes that allow or restrict **relative**	Closest shell-to-shell *actuation* mechanism. Not a twist-lock and not a collapsing mechanism (it is rotational-impact / MIPS), but an examiner could cite it for "user-actuated relative

Ref	Title / Assignee	What it discloses	Impact on twist claim
		sliding of the shells	movement between shell components"
CA2944358A1 / EP3125712A1	Locking liner for helmet	Liner lock between shell and liner	Liner locking concept — no twist
US9554611B2	Foldable protective helmet (Arrouart)	Foldable shell, commercial folding lineage	Background art
WO2018052940A1 / US10959480B2 / US11678711	Machine-vendible foldable bicycle helmet	Foldable honeycomb-matrix bicycle helmet family (antecedent of the US12268268/ US12740611 "expandable" line)	Reinforces anticipation of elements 1–5
CA3096994A1	Adjustable helmet assembly	Locking mechanism = **clamp / lever / cam** function; classified **A42B 3/32**	Shows cam/ lever locking in the collapsible-helmet class

CPC class confirmed

- **A42B 3/32** — *Collapsible helmets; Helmets made of separable parts; Helmets with movable parts, e.g. adjustable*
- **A42B 3/328** — *. . with means to facilitate removal, e.g. …*

These are the correct primary classes for a full attorney-led search.

US11857020 — re-read (the rotation reference)

Claim 1 confirms: an impact-resistant **base body** + an impact-resistant **crown**, **coupling means** allowing the crown to move **relative to the base body** between a protective position and a

stowed position, with **releasable locking means** locking the two together in each position, and the description explicitly uses **rotation of the crown relative to the base body**. This is **rotation between two helmet components + releasable lock** — the exact functional genus of the twist claim, differing only in the *degree* of rotation (full rotation/stow vs. a 90° twist) and in the structure being a honeycomb liner.

Verdict on the twist claim (unchanged, now better supported)

No patent was found that discloses a **quarter-turn (≈90°) twist of an outer shell relative to an inner honeycomb liner to switch the liner between a rigid expanded state and a flat collapsed state.**

However, the surviving claim is narrow and faces a strong § 103 (obviousness) posture:

1. **US11857020** already teaches *rotation between helmet components + releasable locking* as the collapse/stow actuation.
2. **US11944149 (MIPS)** teaches a *user-operable switch* that toggles the *locked / sliding* state between an inner and an outer shell.
3. **Quarter-turn / bayonet twist-locks** are ubiquitous mechanical art (US4893978, US6575682, US9200660, US5123795, US4653970 — already cited in v3).
4. Substituting a known quarter-turn fastener for the known rotation-and-lock of US11857020, applied to the known honeycomb-collapsing helmet of WO2023233224A1, is a textbook *KSR v. Teleflex* combination.

Bottom line: the twist actuation is the only element not literally read on by a single reference, but it is very likely obvious in view of the above. Utility-patent protection here is a long shot. The more defensible paths remain a **design patent** on the visual appearance and/or a **provisional** to hold a priority date while an attorney runs the A42B 3/32 classification search.

Search date: 2026-10-09 · Sources: Hermes web_search (Google Patents, Justia, FPO, Exa legal, EPO data), OpenAlex/Crossref